

# Manoj Kumar

## Principal Software Engineer

Seasoned **Principal Software Engineer** with **20+ years of experience** in designing and scaling **distributed systems, cloud-native architectures, and enterprise SaaS platforms**. Proven expertise in **software engineering, cloud service management, and architectural leadership**, driving strategic initiatives and delivering highly scalable, customer-centric solutions. Adept at leading cross-divisional teams, mentoring senior engineers, and fostering a culture of innovation and technical excellence. Recently, I have been focusing on Cloud Native Application

Throughout my career at **Fidelity Investment, Dell Technologies, and Ford Motor Company**, I have tackled complex challenges such as **Edge computing, BareMetal Provisioning platform, Data pipeline scalability and Serverless resource allocation**. Passionate about leveraging **cutting-edge technologies** to solve intricate business problems, always with a **customer-first mindset**.

## WORK EXPERIENCE

### Fidelity Investments, Durham NC — *Principal Software Engineer*

June 2024 – Present

- Architected and Designed Swagger-Powered/Schema Aware AI assistant Test with LangChain and LLM, and VS Code
- **Intern Leap Project:** Led the end-to-end delivery of the LEAP TLM Pro project—a Maven-based plugin with Jira, Snowflake integration—to proactively manage technology lifecycle (EOL libraries and certificates), streamline remediation, reduce deployments, and enhance cross-team collaboration
- **Modernized Legacy System** Led the re-architecture of MM226 Reconciliation Re-Platform, transforming a legacy batch system into a scalable, serverless AWS solution.
- **Built Resilient Architecture** Designed an active-active deployment across AWS us-east-1 and us-east-2 for high availability and fault tolerance.
- **Orchestrated Cloud Workflows** Implemented data reconciliation using AWS Step Functions, Lambda, S3, DynamoDB, and AWS Batch.
- **Engineered Secure APIs** Developed Spring Boot-based RESTful APIs for trade break management and reconciliation, with robust security and data access controls.
- **Enabled Intelligent Ingestion** Integrated schema validation and business rule enforcement at ingestion points for TA and mainframe feeds using Lambda's AWS Batch and Spring Batch.
- **Led and Mentored Teams** Served as hands-on Tech Lead, mentoring developers and aligning with users and enterprise goals for cloud modernization.
- Architected and design PIP fund merger process using AWS serverless with Spring batch.
- Established GitHub branching strategies and integrated LaunchDarkly for feature flag management and progressive delivery

## CONTACT

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Raleigh, North Carolina

## ACHIEVEMENTS

- Dell Patent for Unified API
- Impact Award-Dell
- Game Changer Award-Dell
- Impact Award-Fidelity

## SKILLS

### Professional

- Cloud Computing, Distributed Systems
- Microservices Architecture
- API Design/Development
- Platform Engineering,
- Software Development
- REST/gRPC API, Agentic AI

### Technical

- Java / Spring Boot
- Spring Eco System
- Oracle/Postgres, NoSQL (MongoDB)
- Messaging Queues (Kafka / RabbitMQ)
- Kubernetes and Docker
- AWS Tech Stack
- LangChain/LangGraph/RAG
- AWS Bedrock

## Certifications

- AWS AI Practitioner
- Generative AI
- AWS Solution Architect Associate
- AWS Developer Associate
- Pivotal Cloud Foundry (PCF) Developer

## Blogs

- [GitHub Copilot](#)
- [AI-Driven Automation](#)
- [Component Test with LangChain](#)

## **Dell Technologies, Seattle WA — Senior Principal Software Engineer**

Oct 2022 – April 2024

- Architected and led the development of a **BareMetal Platform** for managing lifecycle operations of physical (bare metal) servers, including **onboarding, offboarding, and telemetry data collection**.
- Designed and implemented **high-performance REST and gRPC APIs** enable scalable service integrations across internal systems.
- Integrated Redis for low-latency caching and MongoDB for telemetry and metadata storage, optimizing system responsiveness and reliability.
- Established **component-level automated tests using REST-assured**, improving regression coverage and early defect detection.
- Delivered a unified control plane to handle **device provisioning, firmware updates**, and real-time telemetry streaming for fleet-level bare metal systems.
- Played a key role in improving **onboarding time of new hardware** by 40% through workflow optimization and telemetry feedback integration.
- Provided leadership in developing long-term architectural plans for product development
- Led the Dell Apex cloud wide initiative for Unified API for streamline Dell Private and Public API

## **Ford Motor, Dearborn, MI — Sr. Technical Anchor**

Dec 2021 - Oct 2022

- Conducted research and evaluation of emerging **materials, components, and technologies** to enhance product design and technical innovation.
- Collaborated with business stakeholders to **gather and analyze requirements** for new features and enhancements to cloud-native applications.
- Designed and developed **scalable, cloud-native applications and services** aligned with business goals and performance standards.
- Architected technical solutions by defining **end-to-end system architecture**, ensuring compliance with scalability, security, and maintainability best practices.
- Developed RESTful APIs using **Spring Boot, Spring Data JPA, and Spring Cloud**, facilitating microservices communication.
- Provided hands-on technical support to the team through **troubleshooting, peer code reviews, and integration testing**.
- Resolved application issues raised by stakeholders, performing root cause analysis and applying targeted fixes.
- Designed secure API contracts in compliance with 42Crunch security standards, ensuring robust protection of data interfaces.
- Implemented Spring **Security with OAuth and ADFS integration** to enable secure authentication and authorization mechanisms.
- Built and maintained **CI/CD pipelines** using **Jenkins DSL**, automating deployment and testing workflows for improved delivery velocity.

## **World Fuel Service Miami, Florida — Principal Engineer**

Nov 2019 - Dec 2020

- Analyzed business requirements, designed application architecture, and implemented scalable solutions for cloud-native systems.
- Authored **Architecture Decision Records (ADRs)** to document key architectural decisions and rationale for future reference
- Developed REST and batch APIs using **Java/Spring Boot** for integration with external systems, including **QuickBooks** and **FAA** data upload services.
- Implemented the lambda function on **AWS API gateway** for authentication. Secured the API's swagger endpoint by implementing Spring boot swagger security. Exposed the endpoints using Spring boot Rest and Spring batch framework for bulk operations like uploading data to the FAA (Federal Aviation Administration).
- Implemented **reply queue messaging patterns** using **AWS SQS** to support asynchronous processing and system decoupling
- Wrote **BDD component tests** using **Cucumber, Spring Boot, and WireMock**, ensuring high test coverage and behavior validation.
- Built a generic **exception handling module** using **@ControllerAdvice** in Spring Boot to standardize business and system error handling.
- Integrated with AWS S3 Service and developed the common module to upload and download the file for better reusability and maintainability.
- Created a reusable **AWS S3 file module** for efficient file upload/download operations, promoting modularity and maintainability across services.

## **Ford Motor Dearborn, Michigan — Lead software Engineer**

Mar 2015 – Nov 2019

- Led the modernization of legacy on-premise architecture by proposing a **target cloud architecture**, aligning with scalability, resilience, and DevSecOps principles.
- Decomposed monolithic business functions into **independent, deployable microservices**, following **12-Factor App design** patterns.

- Architected and developed backend microservices using **Spring Boot, Spring Data, and Spring Cloud**, with secure HTTP/HTTPS communication and JSON-based data exchange.
- Designed and implemented a **Single Sign-On (SSO)** solution using **OAuth 2.0** and **ADFS**, ensuring consistent authentication and authorization across services.
- Configured **Jenkins CI/CD pipelines** for seamless deployment of microservices on **Pivotal Cloud Foundry (PCF)**, streamlining release cycles.
- Implemented **Spring Cloud config server** for keeping the configuration setting out from microservice.
- Integrated **SonarQube** with Jenkins for continuous **code quality analysis**, improving maintainability and reducing technical debt.
- Served as a **technical lead**, guiding engineering best practices, conducting peer reviews, mentoring junior engineers, and promoting a strong DevOps culture.
- Actively participated in Agile development processes including **Pair Programming, Mob Programming, Sprint Planning, Backlog Grooming, and Retrospectives**.
- Collaborated cross-functionally with product owners, architects, and QA teams to align deliverables with business goals and ensure timely, high-quality releases.

### **Offshore Technical Lead – Java/Spring/Webservices (Backend)**

**Bharti Telesoft, Polaris, TCS (Tata Consultancy Services), ITC Infotech**

Duration: 2002-2013

Led end-to-end design and development of enterprise applications using Java, Spring, Hibernate, and Web Services for global clients. Managed cross-functional offshore teams, conducted code reviews, and ensured on-time delivery of high-quality software. Played a key role in system integration, performance tuning, and deployment of scalable applications across telecom, banking, and IT domains. Collaborated closely with onsite counterparts to translate business requirements into robust technical solutions.

## **EDUCATION**

- **Master of Computer Applications (MCA)**, CCS University, Meerut, India
- **Bachelor of Science (B.Sc.)**, DR. B.R Ambedkar University, Agra, India